SPACE X

Predicting Falcon 9 First-Stage Landings

Applied Data Science Capstone • Technical Report

A reproducible pipeline from API and web data to interactive analytics and classification.

Armando Romero Mariscal Guasp

https://github.com/ijontichyai/IBM-Data-Science-Capstone

21 August 2026

Executive summary

Can launch history reveal when a Falcon 9 first stage is likely to land?

90
launch records analyzed

4
launch sites compared

90.4%
highest cross-validation score

83.3%
best holdout accuracy

What we did

- Collected Falcon 9 launch data via API and web scraping
- Cleaned, encoded, queried, mapped, and visualized the data
- Tuned Logistic Regression, SVM, Decision Tree, and KNN

What we learned

- Launch site, payload, orbit, and booster context are associated with landing outcomes
- Decision Tree achieved the highest CV score (90.4%), but Logistic Regression, SVM, and KNN tied for the best holdout accuracy at 83.3%.
- LR, SVM, and KNN each produced 3 false positives and 0 false negatives on the holdout set.

Business context and analytical problem

WHY IT MATTERS

First-stage reuse can reduce launch cost and improve mission economics.

A reliable landing prediction helps planners estimate reuse potential before launch.

Research question

Given launch conditions and historical outcomes, will the first stage land successfully?

INPUTS	Payload • Orbit • Site • Booster • Flight history
TARGET	Class = 1 landed | Class = 0 did not land
OUTPUT	Transparent evidence for decision support

From raw launch records to decision evidence

01	02	03	04	05
COLLECT	**WRANGLE**	**EXPLORE**	**INTERACT**	**PREDICT**
API + Wikipedia	Normalize + label	Plots + SQL	Folium + Dash	GridSearchCV

Design principle

Each analysis stage is traceable to source data, transformations, and a reproducible notebook or Python file in the repository

Data collection — SpaceX REST API

GET
/v4/launches/past

LOOK UP
rockets · payloads · launchpads · cores

NORMALIZE
JSON → pandas DataFrame

FILTER
Falcon 9 launch records

EXPORT
structured CSV

Core fields retained

- Flight number and date
- Booster / core configuration
- Payload mass and orbit
- Launch site and landing outcome

Quality controls

- Resolve related IDs before analysis
- Filter irrelevant records and inspect missing values
- Preserve a repeatable extraction path
- Store the final analytical table

Data collection — web scraping

1. Request static Wikipedia launch-history snapshot
2. Parse HTML tables with BeautifulSoup
3. Extract rows and normalize cell text
4. Build a structured launch-record DataFrame
5. Validate columns and export for analysis

Fields extracted

- Flight No.
- Launch site
- Payload / orbit
- Customer
- Launch outcome
- Booster landing

Data wrangling and feature engineering

CLEAN

Impute missing payload mass with the mean and inspect remaining missing values

LABEL

Convert landing outcome to binary class 0 / 1

ENCODE

One-hot encode orbit, site, landing pad, and serial

SCALE

Standardize the feature matrix before model training

Result: one analysis-ready feature matrix X and one target vector Y.

Exploratory analysis — questions before models

SITE

Where do launches succeed most often?

PAYLOAD

How does mass relate to outcome?

ORBIT

Which mission profiles are most successful?

BOOSTER

Which configurations perform best?

TIME

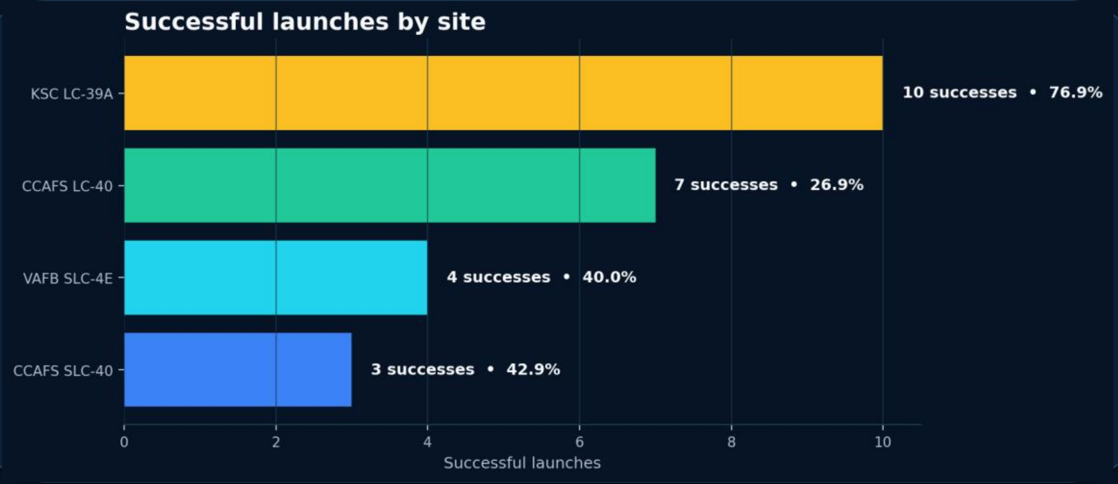
Does success improve as experience grows?

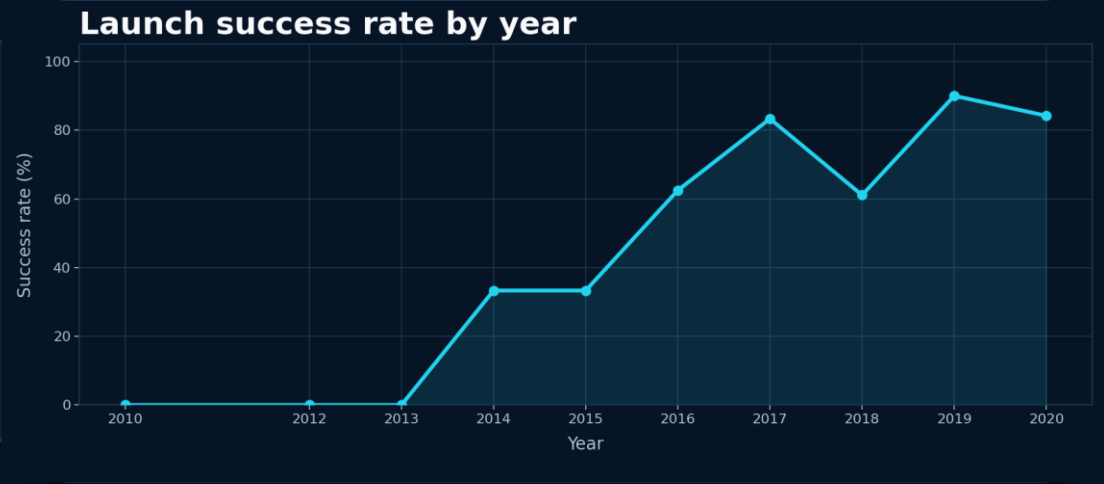
RISK

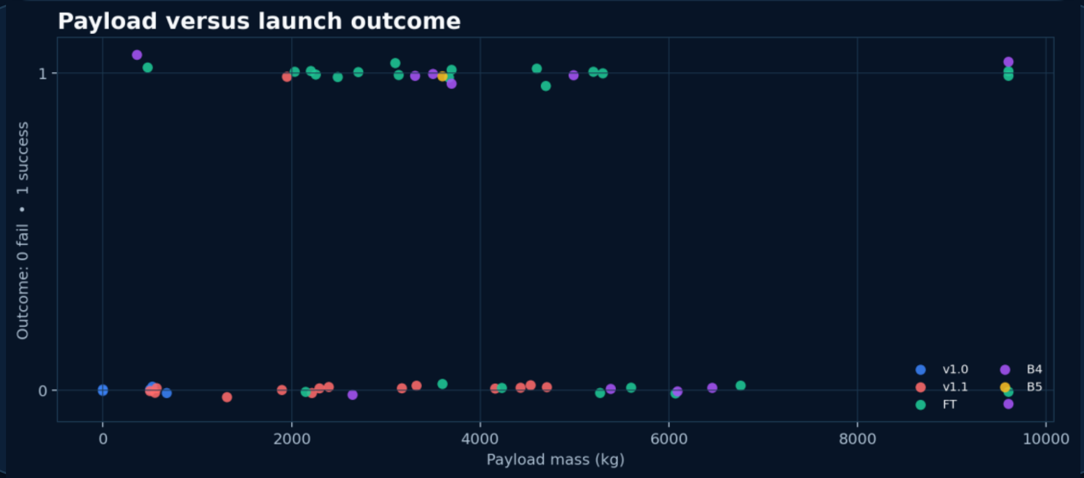
Under which conditions are failures more common?

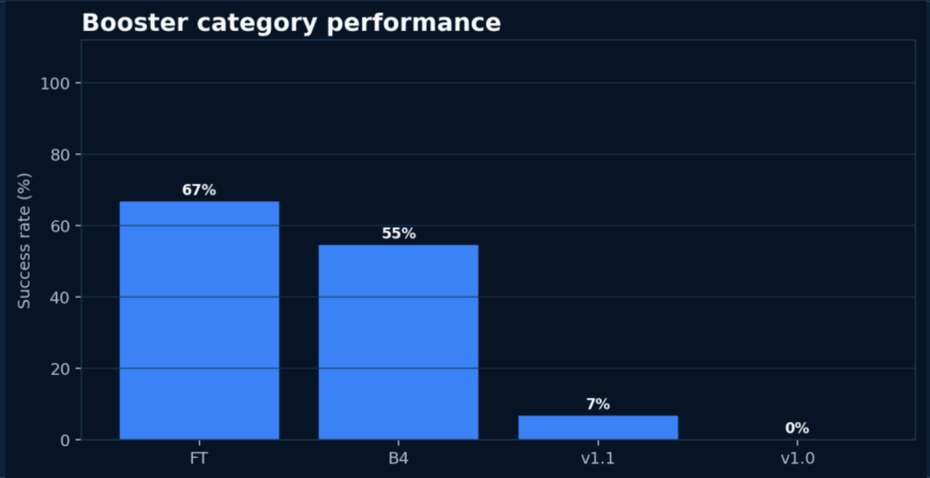
Bar charts compare site and booster performance; scatter plots examine payload vs. outcome; line charts show yearly success trends; SQL verifies exact aggregations.

EDA results — site, payload, and experience









EDA with SQL — reproducible analytical questions

01	SITE COVERAGE	DISTINCT launch sites and CCA-prefixed records
02	PAYLOAD	SUM and AVG payload mass by customer / booster
03	SUCCESS	COUNT outcomes and drone-ship landings
04	RANKING	GROUP BY landing outcome and ORDER BY count
05	TIME	Filter launch dates and compare historical periods

Typical pattern: SELECT → WHERE → GROUP BY → SUM / AVG / COUNT → ORDER BY

SQL results — exact summaries support the story

QUESTION	RESULT	USE
Launch sites	4 unique launch sites identified	Coverage
NASA (CRS) total payload	45,596 kg total payload for NASA (CRS)	Aggregation
F9 v1.1 average payload	≈2,928 kg average payload (F9 v1.1)	Aggregation
First ground-pad landing	First success: 2015-12-22	Time
Landing-outcome ranking	No attempt 10; drone-ship failure 5; drone-ship success 5	Ranking

Interactive geospatial analytics with Folium

MAP LAYERS

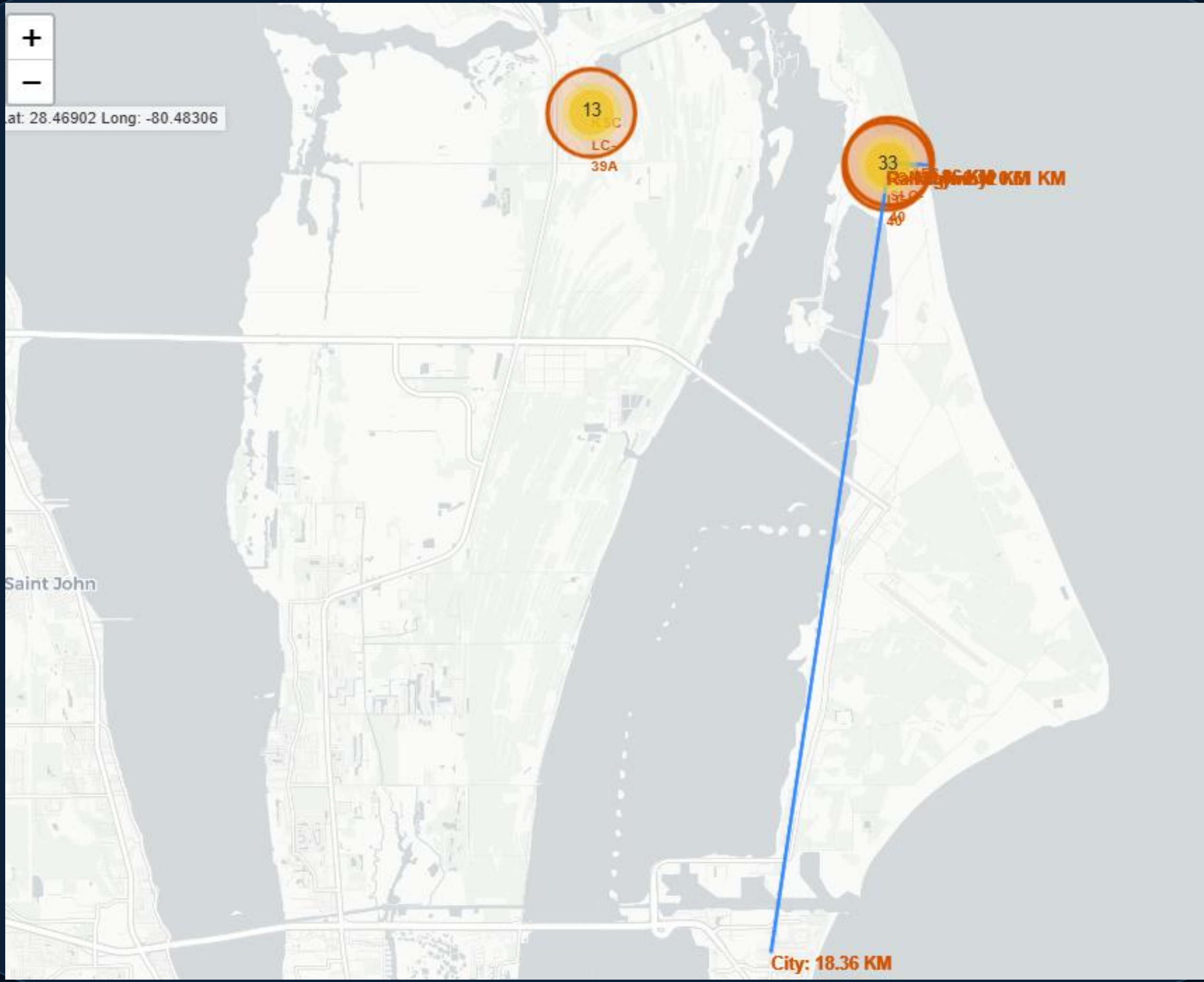
1	**Launch sites**	Circles + labels
2	**Launch records**	Green success / red failure markers
3	**Clusters**	MarkerCluster for overlapping launches
4	**Proximity**	MousePosition + distance lines

PROXIMITY TESTS

- **Coastline**
- **Highway**
- **Railway**
- **City / population center**

Goal: explore geographic patterns around launch facilities.

Folium results — CCAFS SLC-40 proximity analysis



Key geographic findings

Measured proximity from CCAFS SLC-40

- Railway: 0.12 km
- Highway: 0.61 km
- Coastline: 0.86 km
- City: 18.36 km

Plotly Dash — turning analysis into an interface

INPUTS

- Launch-site dropdown
- Payload range slider

USER ACTION

Select a site and constrain payload mass.

CALLBACKS

Site dropdown → success pie chart

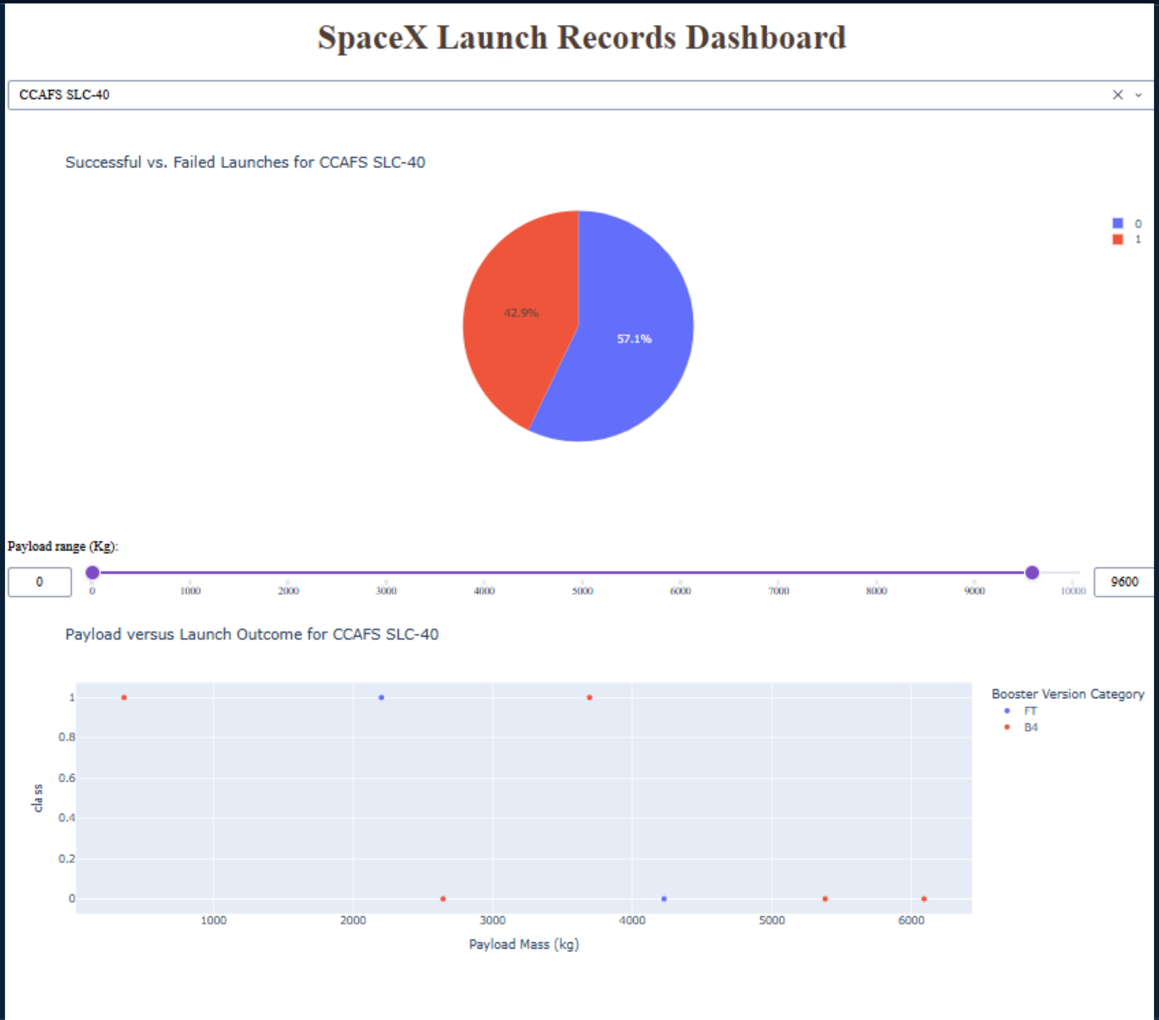
Site dropdown + payload slider → payload scatter plot

OUTPUTS

- Success pie chart
- Payload scatter plot

Figures update interactively when relevant inputs change.

Dash results — site selection and payload filtering



Observed insights

42.9%

CCAFS SLC-40 success rate

3 of 7

launches at CCAFS SLC-40 were successful

The payload slider filters the scatter plot; the site dropdown updates both visuals through callbacks.

Predictive analysis — model training and comparison

SPLIT
80% train / 20% test
random_state = 2

STANDARDIZE
Fit scaling on X
reassign transformed X

TUNE
GridSearchCV
10-fold CV

EVALUATE
Holdout accuracy
confusion matrix

Models evaluated

- **Logistic Regression** — C · penalty · solver
- **Support Vector Machine** — C · gamma · kernel
- **Decision Tree** — criterion · depth · splitter · features · samples
- **K-Nearest Neighbors** — neighbors · algorithm · p

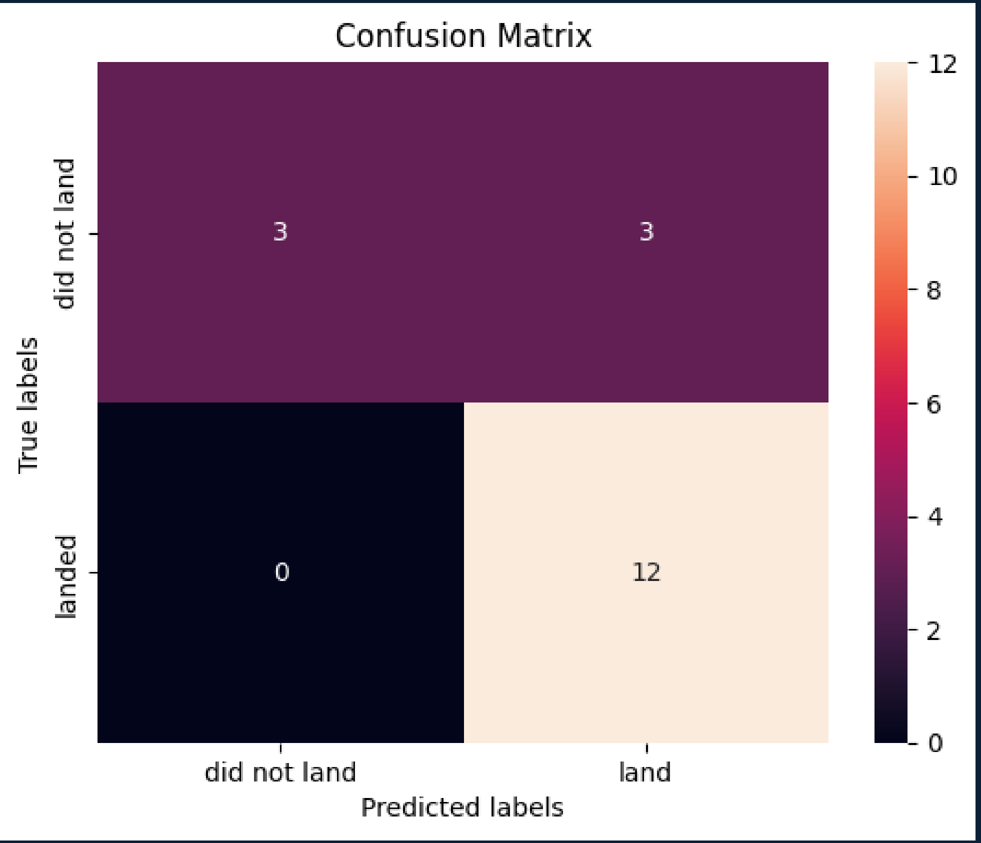
Model results — cross-validation vs. holdout performance

Verified accuracy

Model	CV	Test
Logistic Regression	CV 84.6%	Test 83.3%
SVM — sigmoid	CV 84.8%	Test 83.3%
Decision Tree	CV 90.4%	Test 66.7%
KNN	CV 84.8%	Test 83.3%

Decision Tree achieved the highest CV score (90.4%), while Logistic Regression, SVM, and KNN tied for the best holdout accuracy at 83.3%.

Confusion matrix — LR / SVM / KNN



Conclusions and innovative next steps

What the evidence supports

- Launch outcomes improve across the flight sequence
- Decision Tree achieved the highest cross-validation score at 90.4%
- Logistic Regression, SVM, and KNN tied for the best holdout accuracy at 83.3%
- LR, SVM, and KNN each produced 3 false positives and 0 false negatives

Beyond the template

1 COST-AWARE THRESHOLD
Penalize false positives more heavily than false negatives.

2 CALIBRATED PROBABILITY
Report landing probability—not only a binary class.

3 DRIFT MONITORING
Retrain as new booster versions and mission profiles appear.

Recommendation: use as decision support, not an autonomous go/no-go rule.